

VR and Virtual Environments

A man and a woman are wearing VR headsets and looking upwards in a museum setting. In the background, a large, detailed skeleton of a prehistoric animal is mounted on a stand. The museum has high ceilings with ornate metalwork and various exhibits on display.

Sunoikisis Digital Classics 2025–26: Digital Approaches to Cultural Heritage

19 March 2026

Juliette Quatre (University of London) & Dr Lynn Verschuren (University of Glasgow)



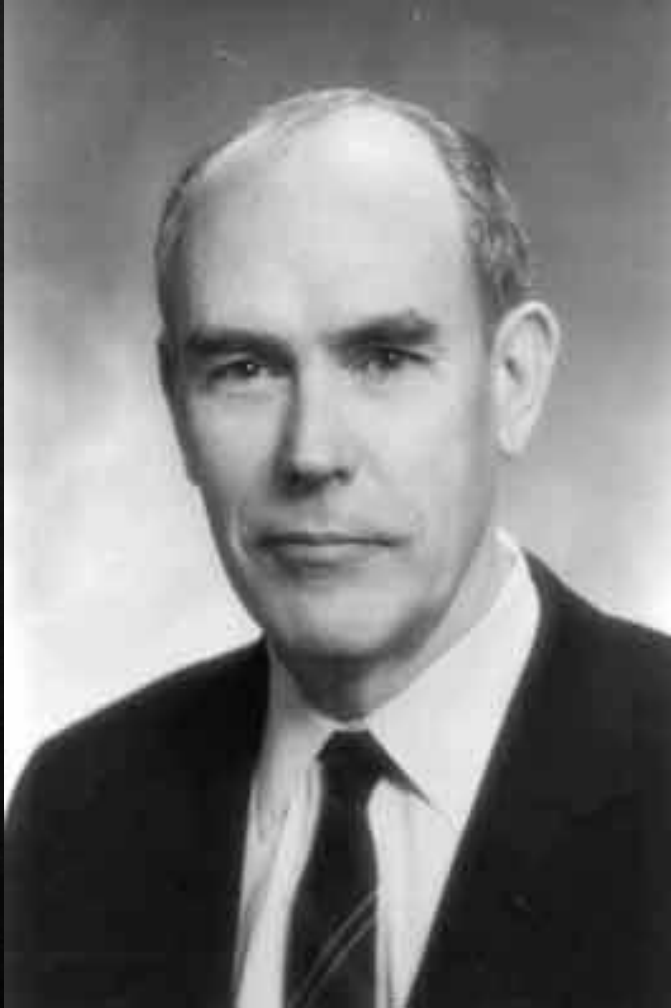
Session Overview

- ◆ Introduction to Virtual Reality
- ◆ VR for Cultural Heritage Collections & Sites
 - ◆ Theoretical Approaches
 - ◆ Case Study 1: The Sarapeion C on Delos
 - ◆ Case Study 2: the Swedish Pompeii Project
 - ◆ Case Study 3: Iconem
 - ◆ Case Study 4: Museums in the Metaverse
- ◆ Discussion & Questions

A marble bust of a bearded man, likely a classical figure, is shown wearing a modern VR headset. The bust is positioned on a white pedestal in a museum setting, with blurred art pieces and columns in the background. The text "Virtual Reality (VR)" is overlaid on the image in a white serif font.

Virtual Reality (VR)

Pioneers: Jaron Lanier



Pioneers: Ivan Sutherland









VR: State of the Art



2016: affordable high-quality VR arrives

Oculus Rift S £300, Vive £500, Odyssey £500, Vive Pro £1,100

OK: needs £1k PC, £2k Laptop

Puzzle: Why do so few people have them?



2018: Standalone 3DoF headsets, early passthrough

Standalone headsets (Oculus GO) need no PC, no infrastructure.

But: Limited 3DoF experience

Passthrough allows us to see fuzzy representation of real world in black and white while onboarding



2025: Standalone 6DoF HMDs

Oculus Quest 3/S (£500/£300) defines the genre: no PC, no wires, no sensors but still 6DoF and full-colour passthrough.

Graphical trade-off, experience still excellent

Next: Better graphics/ergonomics

Medium term: 5G + Edge rendering

Long term: replaced by full-range XR devices that can be worn all-day

VR Features



Computer generated
environments



Sense of depth and 3D



Immersive



Isolating

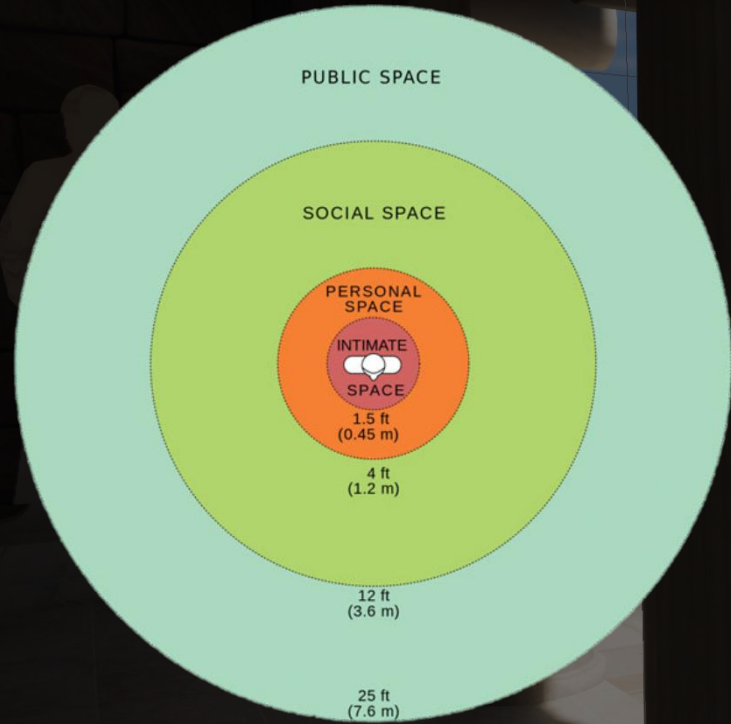


Disorienting

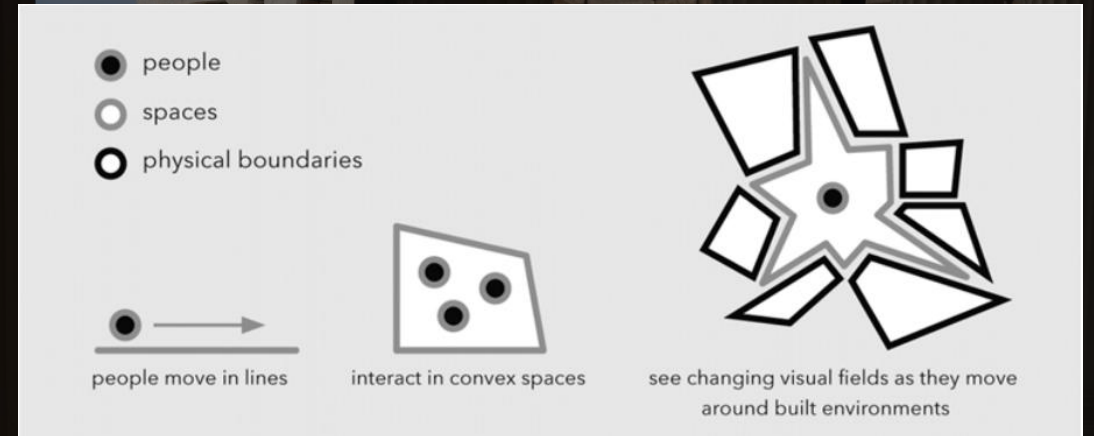
A marble bust of a bearded man, likely a classical figure, is displayed on a white pedestal in a museum setting. The bust is wearing a modern VR headset, creating a juxtaposition of ancient art and contemporary technology. The background shows a museum gallery with other exhibits and warm lighting.

VR & Cultural Heritage

Theoretical approaches in archaeology: experiencing ancient spaces

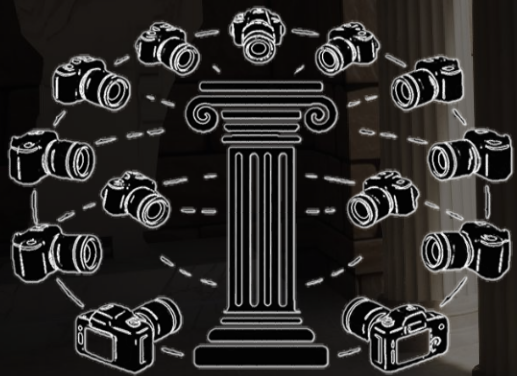


Hall's proxemics spaces as 'reaction bubbles' (Paliou et al. 2015: 125).



The structures of space according to human movements (Paliou et al. 2015: 20).

VR in Archaeology: between interpretation and experience



Photogrammetry

- Metashape
- Manual or automatic
- Capture every angle of an object or building

3D reconstruction

- Sketchup, Blender, etc.
- Manual reconstruction from data and interpretation
- Risk of speculation or errors



VR in Archaeology: between interpretation and experience

Data collection

- Data collected in the field (Raw data): GIS, plans, photographs, drawings, measurements, etc.
- Publications (fieldwork, conference proceedings, books, articles, etc.).

Interpretation

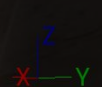
- Main risks: speculation and/or errors
- But investigation and combination of evidence, experience and knowledge of larger trends.

Modelling

- Specialised software: Sketchup, Blender, Metashape, etc.
- Different types of models: photogrammetry, reconstruction, partial photogrammetry .
- Reconstruction of a built environment and its reality

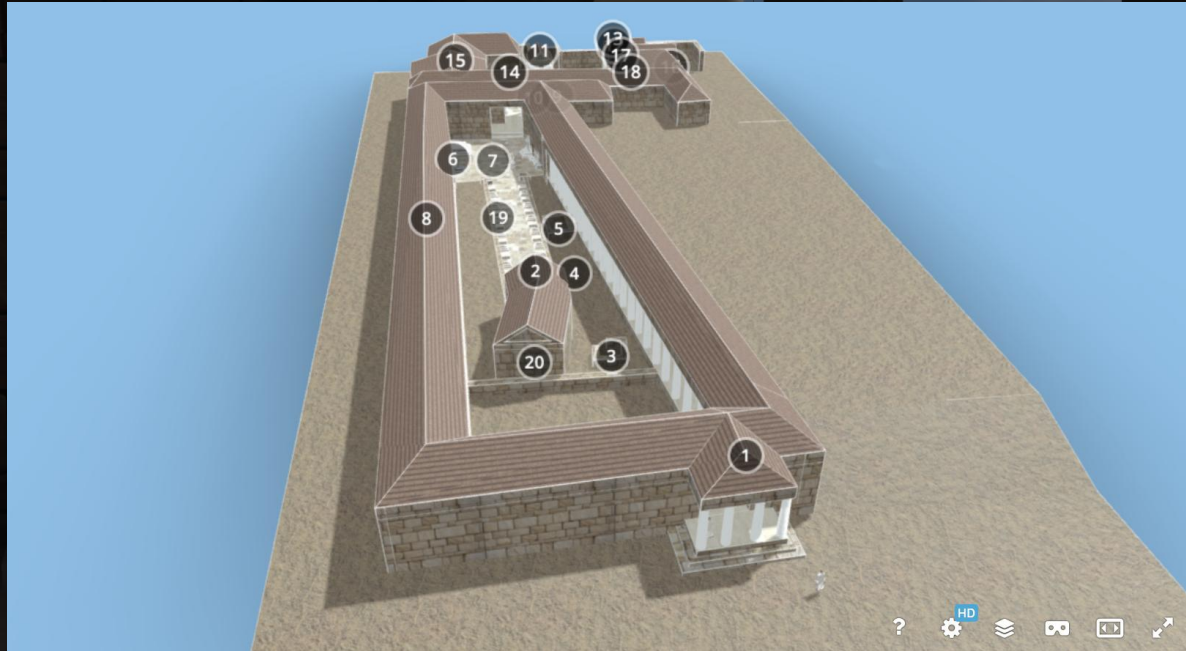
Virtual Reality

- Visualisation.
- More interactive and autonomous approach.
- Accessibility of artefacts or buildings.
- Subjective interactions with the model.
- Teaching resource with different levels of experience.



Case study 1: the Sarapeion C on Delos

Sketchfab



Unreal (game engine)



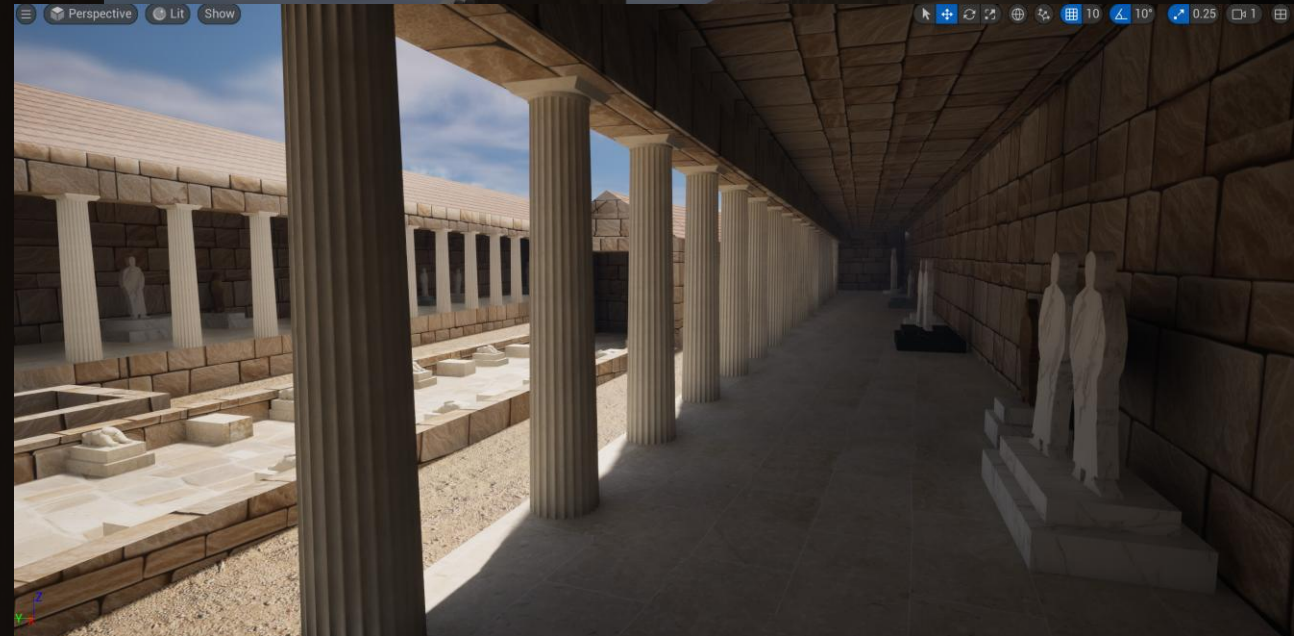
Sarapeion C: bird's eye view from the south



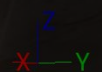
Case study 1: the Sarapeion C on Delos

Sketchfab

Unreal (game engine)



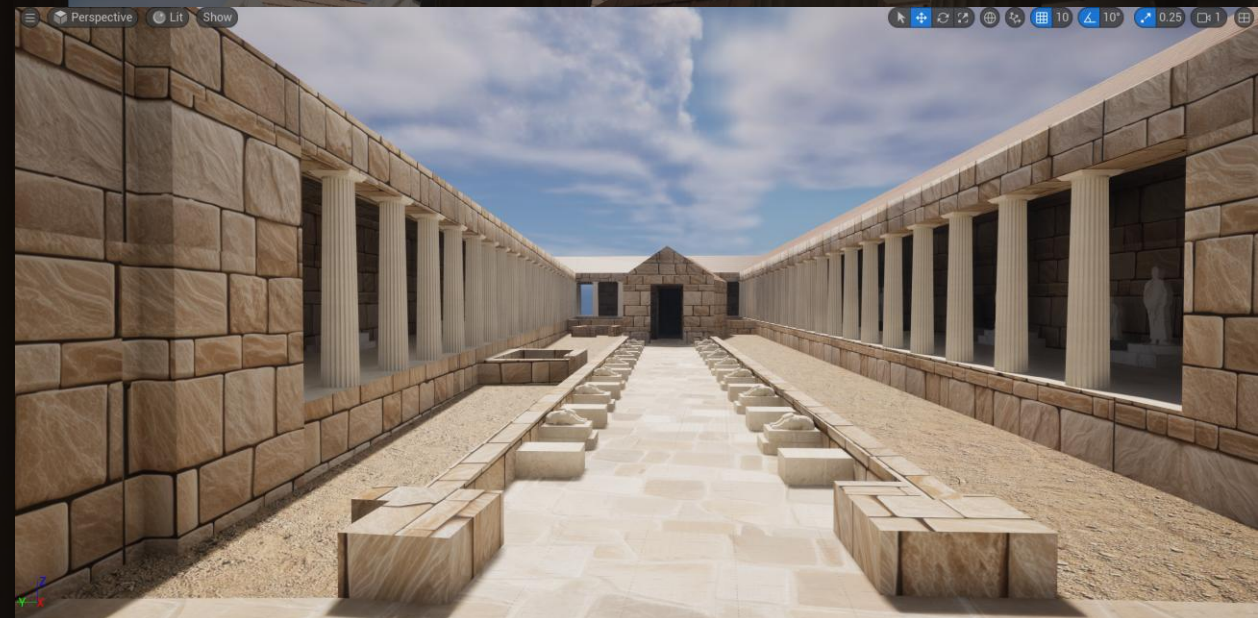
Sarapeion C: view from the east portico



Case study 1: the Sarapeion C on Delos

Sketchfab

Unreal (game engine)



Sarapeion C: the dromos from the north



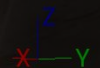
Case study 1: the Sarapeion C on Delos

Sketchfab

Unreal (game engine)



Sarapeion C: view of the northern courtyard and temples
from the L-shaped stoa



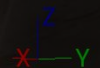
Case study 1: the Sarapeion C on Delos

Sketchfab

Unreal (game engine)



Sarapeion C: view on the northern courtyard
from the temple of Sarapis



Case study 2: the Swedish Pompeii Project

[Lund University](#) [The Joint Faculties of Humanities and Theology](#) [Listen](#)

The Swedish Pompeii Project

The joint faculties of humanities and theology



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[Inscriptions of Insula V 1](#)

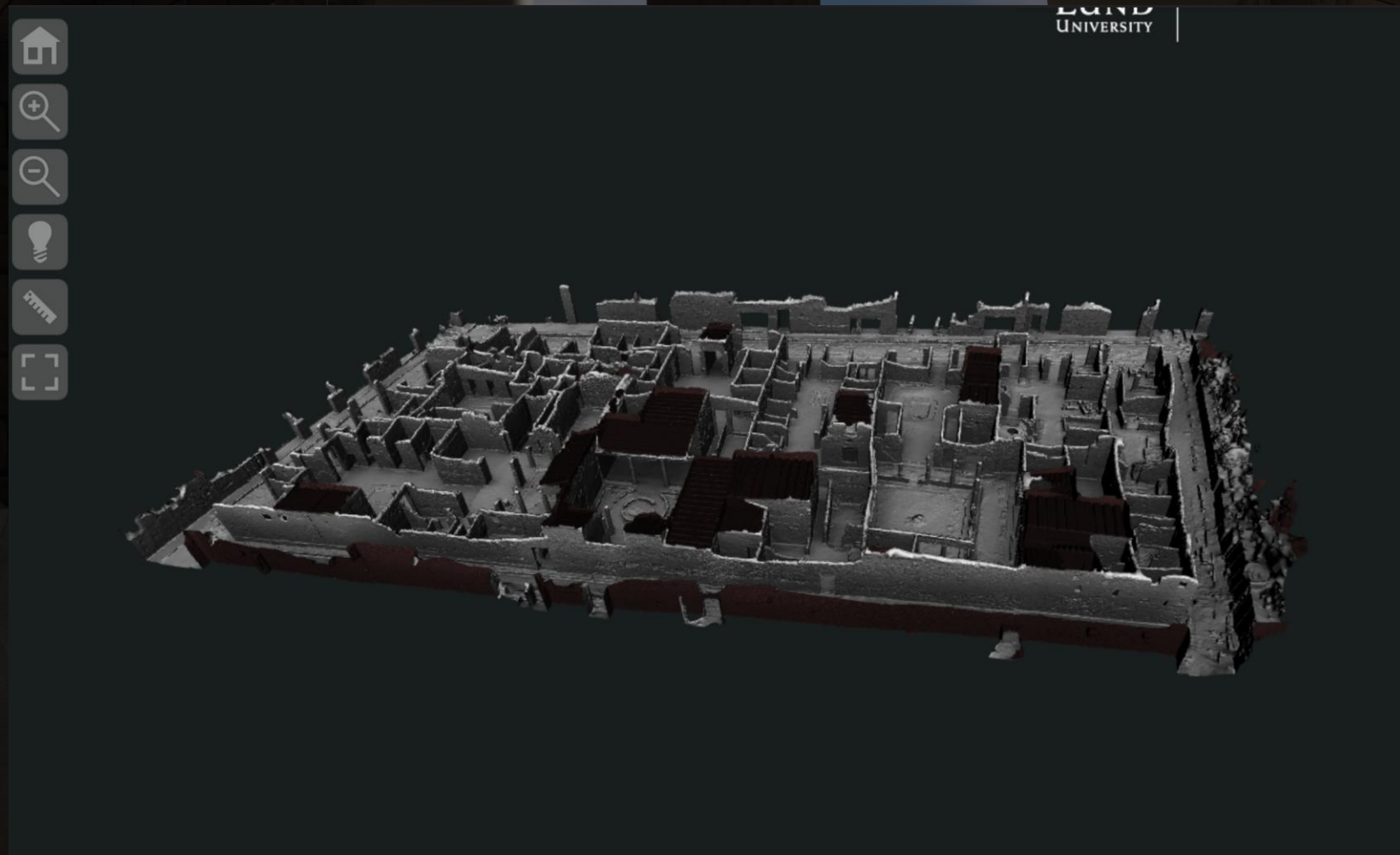
3D Models & 3D GIS



Navigate the 3D-models by selecting a part of the insula. For best possible performance and compatibility, open the link with [Google Chrome](#). (Try to reload the page if the 3D-model does not appear.)



Case study 2: the Swedish Pompeii Project



3D model of Insula V 1 ([Swedish Pompeii Project](#))

Case study 2: the Swedish Pompeii Project



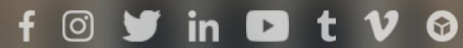
Detailed models of buildings in Insula V 1
([Swedish Pompeii Project](#))



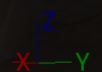
Case study 3: Iconem

iconem
STUDIO

EXPERIENCE
HUMAN
HERITAGE



NOS PROJETS



Case study 3: Iconem - Palmyra



3D reconstruction of the Palmyra archaeological site, showing the Great Colonnade, the Temple of Bel, and the Amphitheatre.

iconem
EXPLORATION

Palmyra (with environment)

STAC ? ↵

DATA LAYERS ≡

MEDIAS (6)

Palmyre-SyrieFacade... Relief_Bel_Baalshami...

Temple_of_Bel_Palm... Palmyra_Monumental...

Palmyra_surrounding... Roman_Empire_200_...

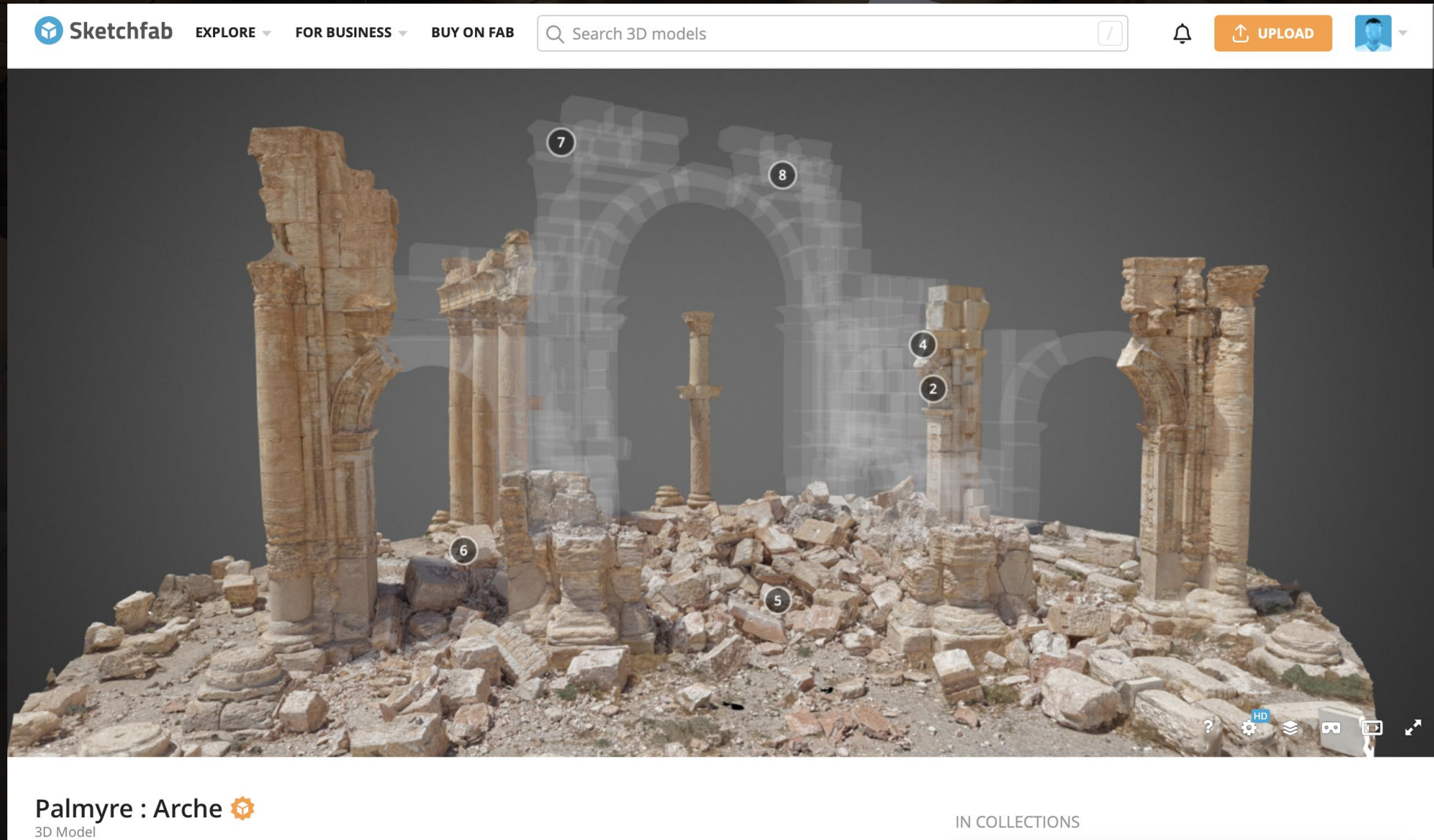
AUDIOS (4)

Palmyra 4 - Great Colonization.wav

Palmyra 3 - Valley of Tombs.wav

Palmyra 2 - Temple of Bel.wav

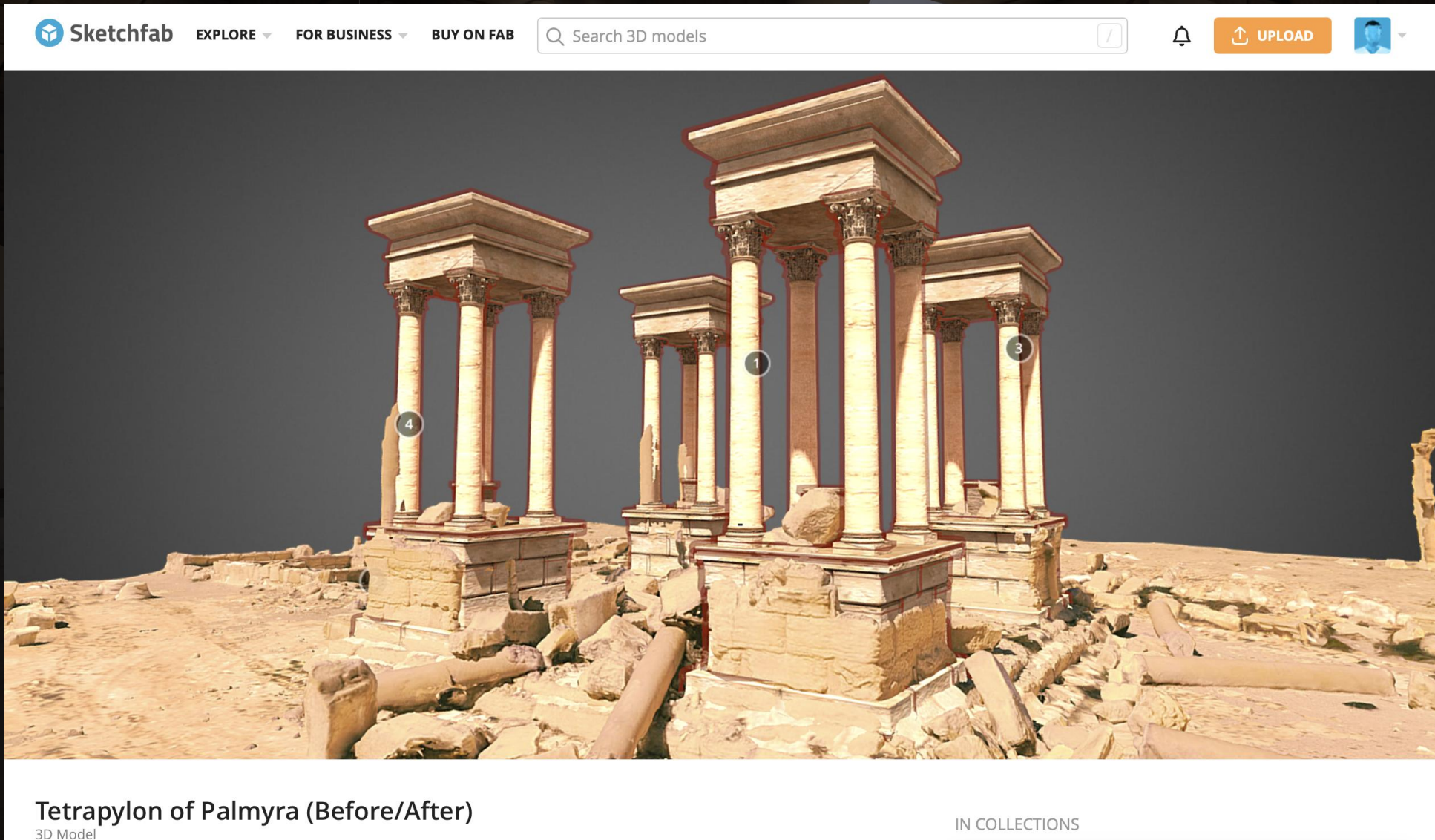
Case study 3: Iconem - Palmyra



Screenshot of [Iconem on Sketchfab](#): Arch in Palmyra with digital anastylosis



Case study 3: Iconem - Palmyra



Tetrapylon of Palmyra (Before/After)

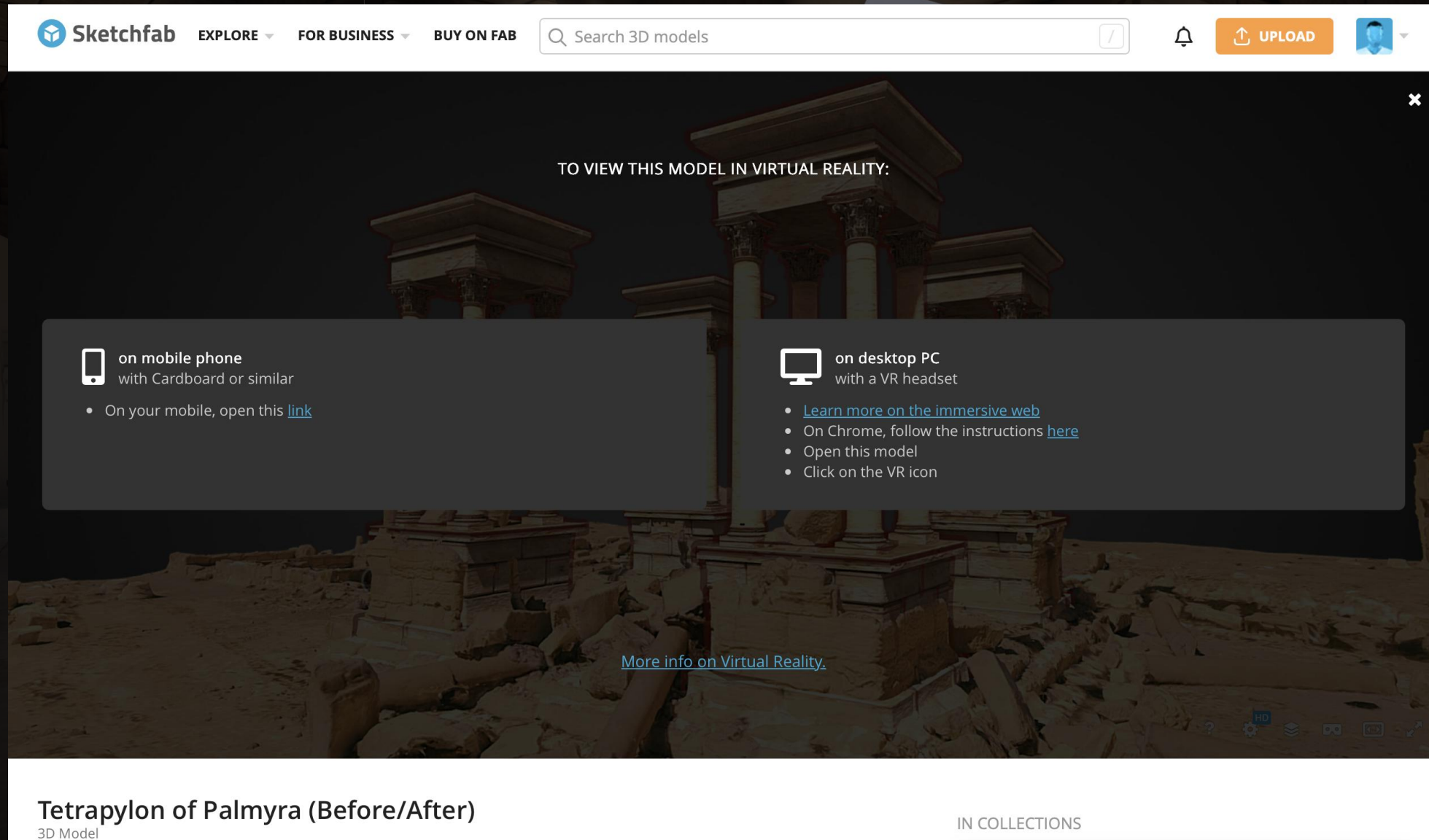
3D Model

IN COLLECTIONS

Screenshot of [Iconem on Sketchfab](#): 3D model of the Tetrapylon at Palmyra showing the site before and after destruction



Case study 3: Iconem - Palmyra



Screenshot of [Iconem on Sketchfab](#): 3D model of the Tetrapylon at Palmyra showing the site before and after destruction





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Museums in the Metaverse

Democratising XR for Cultural Heritage



UK Research
and Innovation



Museums in the Metaverse

- ◆ £6.4M Levelling Up Innovation Accelerator Project (2023-26)
- ◆ Led by UofG in collaboration w/ Edify, The Hunterian, National Museums Scotland, Historic Environment Scotland (Phase 1)
- Project Aim: To make good on the economic promise of XR for the Cultural Heritage Sector





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Museums

Huge economic and cultural importance, physically constrained.



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Vast majority of objects in storage



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Sensitive Objects



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Audience barriers



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Virtual Museums

Obvious.... yet challenging.



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Technology Barriers





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Economic Barriers





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Control



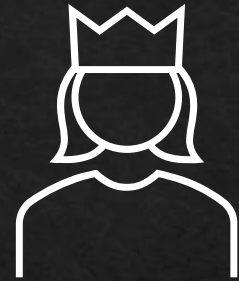


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The MiM Solution

The core of the Museums in the Metaverse project.



No Code Build-Your-Own (BYO) Museum Platform

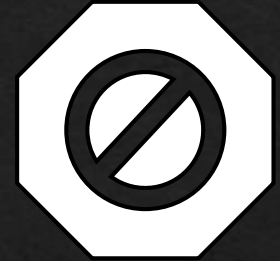
* *Visitors:* Access and Interact with curated MiM experiences on-site or remotely

* *Creators:* Heritage Professional, Lay Creator, Educator...



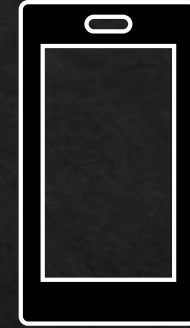
BYO Museum with your own 3D assets or the heritage collections produced by MiM project team

User Generated Content (UGC)



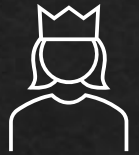
IP, Availability, Veto

Controlled by object owner enabling safe digital loan



Cross-platform

Optimal in VR, available broadly (Desktop & Mobile)



Owners, Platform, Curators
Revenue shared fairly and sustainably

Experience heritage in captivating new ways:

- ◇ + Exquisite Object Detail
- ◇ + Impossible Interactions
- ◇ + Full Immersion
- ◇ + Rich Context(s)
- ◇ Collection-based Research & Teaching
- ◇ International Co-Curation

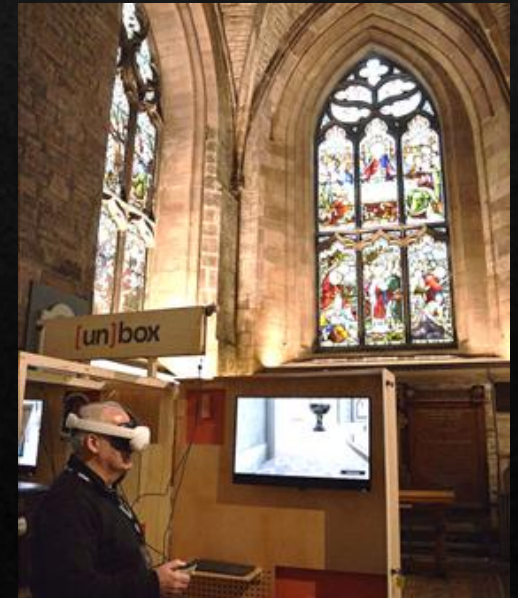




MiM in the Wild: [un]box

[un]box Deployment: Dec 2025 – Mar 2026

- Glasgow Science Centre, Glasgow
 - The Hunterian, Glasgow
 - St Giles' Cathedral, Edinburgh
 - National Portrait Gallery, London
-
- **1,000+ user demos**
 - **500+ in-person surveys**





Tools

 Broadcast

 Stop Broadcasting 00:00:00

 Record Broadcast



A marble bust of a bearded man, likely a classical figure, is shown wearing a modern VR headset. The bust is placed on a white pedestal in a museum setting, with blurred art pieces and columns in the background. The text "Discussion & Questions" is overlaid in a white serif font across the lower half of the image.

Discussion & Questions

References

- Dell 'Unto, N., G. Landeschi, A.-M. Leander Touati, M. Dellepiane, M. Callieri, and D. Ferdani (2016) 'Experiencing Ancient Buildings from a 3D GIS Perspective: A Case Drawn from the Swedish Pompeii Project'. *Journal of Archaeological Method and Theory* 23 (1): 73–94 <https://doi.org/DOI%252010.1007/s10816-014-9226-7>
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- Siard, Hélène. 2009. 'Le Sarapieion C de Délos: Architecture et Cultes'. *Revue Archéologique, Nouvelle Série*, no. 1: 155–61.
- Smith, M., N.S. Walford, and C. Juimenez-Bescos (2019) 'Using 3D Modelling and Game Engine Technologies for Interactive Exploration of Cultural Heritage: an Evaluation of Four Game Engines in Relation to Roman Archaeological Heritage.' *Digital Applications in Archaeology and Cultural Heritage* 14. <https://doi.org/10.1016/j.daach.2019.e00113>

Links

[Swedish Pompeii Project](#)

[Iconem](#)

 [Museums in the Metaverse](#)